6. Vacuum Cleaner Problem

Aim:

To write a Python program to clean two rooms using a simple vacuum cleaner agent.

Algorithm:

1. Initialize the status of both rooms.
2. Check the current room.
3. If dirty, clean it.
4. Move to the next room and repeat.
5. Display the final room status.

Program:

```
rooms = {"A": "Dirty", "B": "Dirty"}
position = "A"
while "Dirty" in rooms. values ():
    print ("Vacuum is in room", position)
    if rooms[position] == "Dirty":
        rooms[position] = "Clean"
        print ("Room", position, "cleaned")
    position = "B" if position == "A" else "A"
print ("Final status:", rooms)
print ("All rooms are clean!")
```

Output:

```
====================== RESTART: D:/AI/vaccum clean
Vacuum is in room A
Room A cleaned
Vacuum is in room B
Room B cleaned
Final status: {'A': 'Clean', 'B': 'Clean'}
All rooms are clean!
```

Result:

Thus, the Vacuum Cleaner problem was implemented successfully using Python.